

Project Proposal: EazyPost – AI-Assisted Social Media Scheduler

1 Project Overview

Small businesses, content creators, and freelancers often struggle to maintain a consistent social media presence across multiple platforms such as Instagram and Facebook. Posting consistently requires time, planning, and creativity.

EazyPost aims to simplify this process by:

- Allowing users to schedule posts in advance.
- Automatically generating captions from ideas using AI if needed.
- Handling multi-platform posting seamlessly via Instagram and Facebook APIs.
- Minimizing user interaction with APIs, tokens, or complex setups.

The goal is to create a **user-friendly SaaS platform** where users can plan, automate, and publish posts efficiently.

2 Key Features

1. User Authentication and Account Connection

- Users can sign up/login via email/password or OAuth.
- Connect their Instagram (Business/Creator) and Facebook Page via Meta OAuth securely.
- Tokens are stored securely in the backend, and users never see raw API keys.

2. Post Scheduling

- Users can schedule posts with media, caption (optional), idea for AI-generated caption (optional), and scheduled date and time.
- Posts are stored in a database for backend processing.

3. AI Caption Generation

- If the user provides an idea but no caption, **EazyPost** uses the Gemini API (or another AI service) to generate an engaging caption automatically.

4. Automated Posting

- **n8n** acts as the automation engine:
 - Watches the database for new or scheduled posts.
 - Checks if the scheduled time is due.
 - Calls AI if needed.
 - Sends post data to the backend for publishing.
- Backend posts content to Instagram and Facebook using the Meta Graph API.

5. Post Status Tracking

- Users can view the status of their posts: Pending, Scheduled, Posted, or Error.
- Backend logs errors and successes for monitoring.

6. Secure Media Handling

- Media files are uploaded via the frontend to cloud storage (S3, Cloudinary, etc.).
- Public URLs are used by n8n/backend to post content.

3 Technical Architecture

3.1 Frontend

- Framework: Next.js / React
- Responsibilities:
 - User authentication and OAuth login
 - Post creation form
 - Dashboard showing scheduled and posted content

3.2 Backend

- Framework: Node.js + Express
- Responsibilities:
 - Securely store access tokens and refresh them when needed
 - Store post data in a database
 - Expose API endpoint `/publishPost` for n8n to call
 - Handle posting to Instagram/Facebook using Graph API

3.3 Automation Layer (n8n)

- Self-hosted instance (e.g., Render)
- Responsibilities:
 - Poll database for scheduled posts
 - Filter posts by scheduled time
 - Call AI API if caption is empty
 - Trigger backend to publish posts

3.4 Database

- MongoDB or PostgreSQL
- Tables/Collections:
 - `users` – stores user info and tokens
 - `scheduled_posts` – stores post metadata, media URL, captions, idea, scheduled time, and status

3.5 Cloud Storage

- AWS S3 / Cloudinary for hosting images/videos
- Ensures that media URLs are publicly accessible for Meta APIs

4 Workflow Example

1. User logs in and connects Instagram/Facebook.
2. User schedules a post: uploads media, provides an idea (optional), sets time.
3. Backend saves the post to the database.
4. n8n detects the new entry and checks if the time is due:
 - If the caption is missing, call Gemini AI to generate it.
5. n8n sends the media + caption + scheduled info to backend `/publishPost`.
6. Backend posts content to Instagram/Facebook using stored OAuth tokens.
7. Status is updated in the database and displayed on the user dashboard.

5 Security & Compliance

- OAuth flow ensures users never share passwords with the system.
- Tokens are encrypted in the database.
- No storing of personal account credentials.
- AI usage is limited to caption generation—no sensitive data is exposed.

6 Advantages

- User-friendly: No API knowledge required.
- Automation: AI-generated captions reduce user effort.
- Multi-platform: Instagram + Facebook integration.
- Scalable: New social platforms can be added later using the same architecture.
- Reliable: Backend handles token refresh, while n8n handles automation.

7 Future Enhancements

- Add TikTok and LinkedIn integration.
- Analytics dashboard: likes, engagement, impressions.
- Auto-suggest hashtags based on post content.
- Multi-user/team accounts with shared dashboards.